

Q1 (2)

$$h = \frac{at^2+a}{2}, k = \frac{2at+0}{2}$$

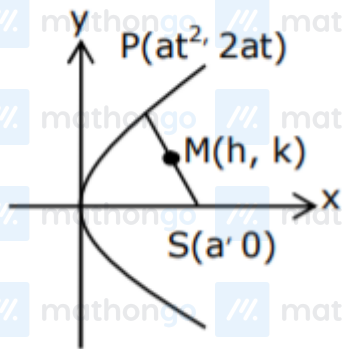
$$\Rightarrow t^2 = \frac{2h-a}{a} \text{ and } t = \frac{k}{a}$$

$$\Rightarrow \frac{k^2}{a^2} = \frac{2h-a}{a}$$

$$\Rightarrow \text{Locus of } (h, k) \text{ is } y^2 = a(2x - a)$$

$$\Rightarrow y^2 = 2a \left(x - \frac{a}{2} \right)$$

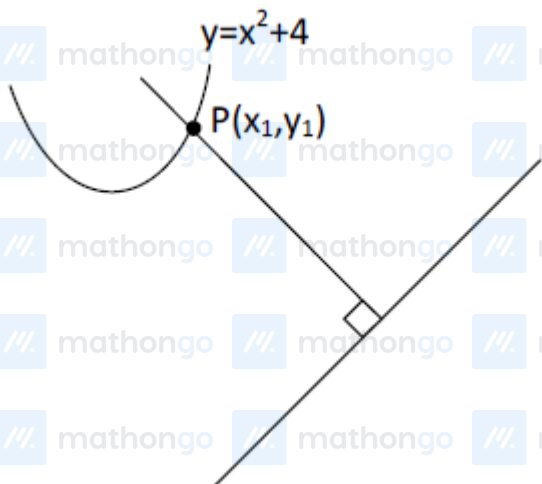
$$\text{Its directrix is } x - \frac{a}{2} = -\frac{a}{2} \Rightarrow x = 0$$



Q2 (4)

$$\left. \frac{dy}{dx} \right|_p = 4$$

$$\therefore 2x_1 = 4$$



$$\Rightarrow x_1 = 2$$

$$\therefore \text{Point will be } (2, 8)$$

Q3 (4)

Equation of tangent : $y = mx + \frac{3}{2m}$ $m_T = \frac{1}{2}$ ($\because$ perpendicular to line $2x + y = 1$)

$\therefore$ tangent is : $y = \frac{x}{2} + 3 \Rightarrow x - 2y + 6 = 0$

Q4 (9)

Circle: $(x - 3)^2 + y^2 = 9$

Parabola: $y^2 = 4x$

Let tangent $y = mx + \frac{a}{m}$

$$y = mx + \frac{1}{m}$$

$$m^2x - my + 1 = 0$$

the above line is also tangent to circle

$$(x - 3)^2 + y^2 = 9$$

$$\therefore \perp \text{ from } (3, 0) = 3$$

$$\left| \frac{3m^2 - 0 + 1}{\sqrt{m^2 + m^4}} \right| = 3$$

$$(3m^2 + 1)^2 = 9(m^2 + m^4)$$

$$6m^2 + 1 + 9m^4 = 9m^2 + 9m^4$$

$$3m^2 = 1$$

$$m = \pm \frac{1}{\sqrt{3}}$$

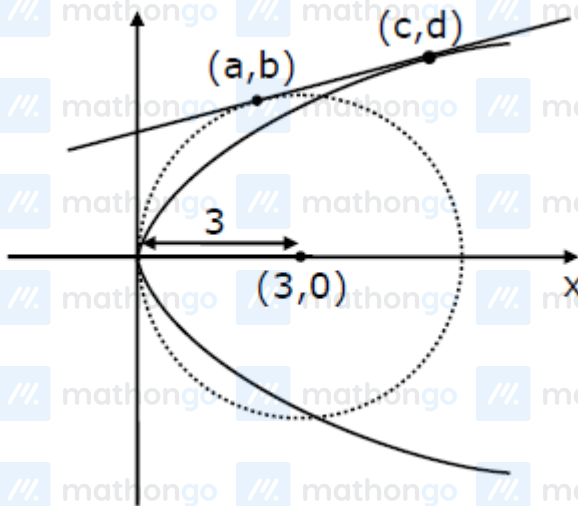
$\therefore$ tangent is

$$y = \frac{1}{\sqrt{3}}x + \sqrt{3}$$

$$\text{or } y = -\frac{1}{\sqrt{3}}x - \sqrt{3}$$

(it will be used) (rejected)

$$m = \frac{1}{\sqrt{3}}$$



for Parabola $\left(\frac{a}{m^2}, \frac{2a}{m}\right) \equiv (3, 2\sqrt{3})$
 (c, d)

for Circle $y = \frac{1}{r}x + \sqrt{3}$ & $(x-3)^2 + y^2 = 9$

$$y = \frac{1}{\sqrt{3}}x + \sqrt{3}$$

$$\text{solving, } (x-3)^2 + \left(\frac{1}{\sqrt{3}}x + \sqrt{3}\right)^2 = 9$$

$$x^2 + 9 - 6x + \frac{1}{3}x^2 + 3 + 2x = 9$$

$$\frac{4}{3}x^2 - 4x + 3 = 0$$

$$4x^2 - 12x + 9 = 0$$

$$4x^2 - 6x - 6x + 9 = 0$$

$$2x(2x-3) - 3(2x-3) = 0$$

$$(2x-3)(2x-3) = 0$$

$$x = \frac{3}{2}$$

$$\therefore y = \frac{1}{\sqrt{3}}\left(\frac{3}{2}\right) + \sqrt{3}$$

$$y = \frac{\sqrt{3}}{2} + \sqrt{3}$$

$$(a, b) \equiv \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$$

$$y = \frac{3\sqrt{3}}{2}$$

$$2(a+c) = 2\left(\frac{3}{2} + 3\right)$$

Hints and Solutions

MathonGo

$$= 2 \left(\frac{3}{2} + \frac{6}{2} \right)$$

$$= 9$$

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

The locus of the mid-point of the line segment joining the focus of the parabola $y^2 = 4ax$ to a moving point of the parabola, is another parabola whose directrix is:

(1) $x = a$

(2) $x = 0$

(3) $x = -\frac{a}{2}$

(4) $x = \frac{a}{2}$

Q2: 24 Feb (Shift 2) - Single Correct

If P is a point on the parabola $y = x^2 + 4$ which is closest to the straight line $y = 4x - 1$, then the co-ordinates of P are :

(1) $(-2, 8)$

(2) $(1, 5)$

(3) $(3, 13)$

(4) $(2, 8)$

Q3: 25 Feb (Shift 1) - Single Correct

A tangent is drawn to the parabola $y^2 = 6x$ which is perpendicular to the line $2x + y = 1$. Which of the following points does NOT lie on it?

(1) $(0, 3)$

(2) $(-6, 0)$

(3) $(4, 5)$

(4) $(5, 4)$

Q4: 25 Feb (Shift 2) - Numerical

A line is a common tangent to the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then $2(a + c)$ is equal to

Answer Key

Q1 (2)

Q2 (4)

Q3 (4)

Q4 (9)